

Update

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Outline

1. Entanglement

- Partial Transpose as Partial Time Reversal

2. CGLMP Test

3. Distributed Quantum Machine Learning

- Dataset Concerns / Issues with MNIST
- Inter-Processor Communication
- Classical Decision Function

4. Compressive QST

5. Post-Selection

- Quantum/Classical Fisher Information
- Separable / Entangled Filter
- Optimal Filter Design

6. DDrf

- Hamiltonian Engineering
- DDrf Spectroscopy
- Analytical Study: Conditional Rotation
- Side-Peak Problem / Apodized Pulses

Entanglement: Reference

The entanglement spectrum of partial transpose in conformal field theory (CFT) was also recently studied. Among other useful numerical methods, tree tensor network, Monte Carlo simulations, and rational interpolations are notable.

Entanglement: Partial Transpose as Partial Time Reversal

For spin systems at fixed spatial locations, partial transposition of the density matrix is unitarily equivalent to partial time reversal, differing only by a local rotation $U_T = e^{-i\pi\sigma_y}$ on the transposed subsystem.

Distributed Quantum Machine Learning: Agenda

1. What data should we use? → Controversial
2. How to capture correlations in data (data embedding)? → Needs further study
3. Inter-Processor Communication → Doable / Straightforward

Distributed Quantum Machine Learning: Dataset Concerns

Our current dataset, originally constructed for a nonlinear classification problem, is not well-defined, may not be physically meaningful, and is not widely used. Moreover, these datasets have well-known solutions, from the perceptron to CNNs.

Distributed Quantum Machine Learning: Dataset Concerns

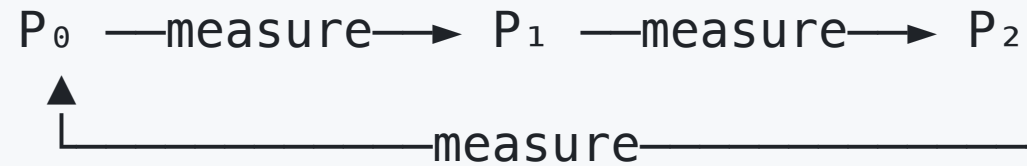
Our current dataset, originally constructed for a nonlinear classification problem, is not well-defined, may not be physically meaningful, and is not widely used. Moreover, these datasets have well-known solutions, from the perceptron to CNNs.

Distributed Quantum Machine Learning: Issues with MNIST

- Due to the input size limitations of quantum simulators, preprocessing MNIST via PCA or resolution reduction fundamentally changes the nature of the learning problem, rendering classical benchmarks meaningless.
- Most studies simplify the task to binary classification between two digits rather than the full 10-class problem, effectively reporting high accuracy on trivially easy tasks that a linear classifier could nearly solve.
- As a result, reported accuracies vary widely from 70% to 99.6% across studies with no consistent interpretation, providing no reliable evidence for the genuine performance of quantum models.

Distributed Quantum Machine Learning: One-Way Cyclic Communication

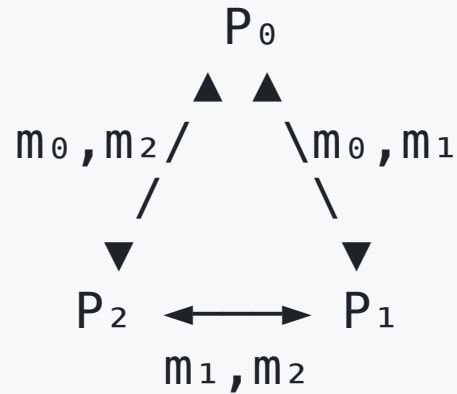
One-way cyclic (`cyclic`):



Each processor measures a qubit and sends the classical outcome to the next processor in the ring. The receiving processor applies conditional gates based on the single measurement outcome.

Distributed Quantum Machine Learning: Joint Cyclic Feedback

Joint cyclic feedback (`multi_fixed` , current default):



Each pair of processors sends their joint measurement outcomes to the third. The decision function $\text{score} = a \cdot m_i + b \cdot m_{\{i+1\}} + c \cdot m_i \cdot m_{\{i+1\}}$ combines both measurements to determine which gate to apply on the target processor.

Distributed Quantum Machine Learning: Classical Decision Function

The cross-processor pooling layer uses a **classical decision function** to determine which quantum gate to apply on a target qubit, based on mid-circuit measurement outcomes from two source processors:

```
score = a · mi + b · m{i+1} + c · mi · m{i+1}

if score >= d:
    apply RX(w0), RZ(w1) to target qubit on processor (i+2)
else:
    apply RX(w2), RZ(w3) to target qubit on processor (i+2)
```

where $m_i \in \{0, 1\}$ is the measurement outcome from processor i , and $m_{\{i+1\}} \in \{0, 1\}$ from processor $(i+1)$.

Distributed Quantum Machine Learning: Boolean Logic Gates

If $m=(m_i, m_{\{i+1\}}) \in \{0, 1\}$, the score function defines a Boolean logic gate. If different (a, b, c, d) settings satisfy the following table, we can implement classical gates:

$m=(0,0)$	$m=(1,0)$	$m=(0,1)$	$m=(1,1)$	Gate
$0 < d$	$a < d$	$b < d$	$a+b+c \geq d$	AND
$0 \geq d$	$a < d$	$b < d$	$a+b+c \geq d$	XNOR
$0 < d$	$a \geq d$	$b \geq d$	$a+b+c \geq d$	OR
$0 < d$	$a \geq d$	$b < d$	$a+b+c \geq d$	COPY

Compressive QST

Post-Selection: Overview

1. Brief Introduction to Measurement Scenario
2. Analytical Approach to Find Maximum QFI
3. Numerical Optimization: Separable Filter
4. Numerical Optimization: Entangled Filter

Post-Selection: Sensing Interaction

Regardless of the quantity being sensed, it requires interaction between the system and the target. In our NV-center system, we perform quantum sensing of the external magnetic field. Such interaction is captured in the Hamiltonian.

$$\mathcal{H}_B = \gamma B S_z$$

where γ is the gyromagnetic ratio, related to magnetic sensitivity. Through this interaction term and time evolution, $e^{-i\mathcal{H}}$, the information on the B -field is embedded in the state.

$$|\psi\rangle \rightarrow e^{-i\mathcal{H}_B}|\psi\rangle.$$

Post-Selection: Quantum Fisher Information

In this sense, the quantum Fisher information (QFI) quantifies how rapidly a quantum state, represented by a density matrix, changes with respect to the B -field. The faster the change, the greater the sensitivity.

$$F_Q = 2 \sum_{k,l} \frac{(\lambda_k - \lambda_l)^2}{\lambda_k + \lambda_l} |\langle k | \mathcal{H}_B | l \rangle|$$

where $\rho = \sum_k \lambda_k |k\rangle \langle k|$.

Post-Selection: Classical Fisher Information

Information about the quantum system is obtained through measurement. That is, the quantum state is projected into a probability distribution $\{p_i\}$.

In statistics, the log-likelihood indicates how plausible it is that B is the true parameter given the observed data.

$$L(B) = \sum_i \ln p_i$$

We can examine how sensitive the log-likelihood is to a small change in the parameter δB . The variance of this score across different data samples tells us how rapidly the slope fluctuates around the peak.

Post-Selection: Classical Fisher Information

- If the variance is large, the slope changes rapidly around the maximum. A sharp peak means that even a tiny shift away from the true parameter θ causes a large drop in the likelihood of observing the data.
- Otherwise, the log-likelihood function is flat and wide. A flat log-likelihood means that many different values of θ are roughly equally likely to have produced the observed data.

In this sense, the classical Fisher information (CFI) is a measure of how rapidly a probability distribution changes with respect to B .

$$F_C = \sum_i p_i \left(\frac{\partial}{\partial B} \ln p_i \right)^2$$

Note that $F_C < F_Q$.

Post-Selection: QFI in a Two-Qubit System

Here is how to calculate QFI F_Q in a two-qubit system. For a pure state $\rho = |\psi\rangle\langle\psi|$, the change in the density matrix with respect to the parameter B is

$$\frac{\partial \rho}{\partial B} = |\partial_B \psi\rangle\langle\psi| + |\psi\rangle\langle\partial_B \psi|$$

Since the state evolves as $|\psi\rangle = e^{-i\mathcal{H}t}|\psi_0\rangle$, the derivative with respect to B yields

$$|\partial_B \psi\rangle = \frac{\partial}{\partial B} e^{-i\mathcal{H}t} |\psi_0\rangle = -iGt|\psi\rangle$$

where $G = \frac{\partial}{\partial B} \mathcal{H}$ is the generator of the parameter shift. The symmetric logarithmic derivative (SLD) $\mathcal{L}$, defined by $\frac{\partial \rho}{\partial B} = \frac{1}{2}(\mathcal{L}\rho + \rho\mathcal{L})$, for a pure state simplifies to

$$\mathcal{L} = 2it_s [\rho, G].$$

Post-Selection: QFI in a Two-Qubit System

The QFI is then obtained as $F_Q = \text{Tr}(\rho \mathcal{L}^2)$, which for a pure state reduces to

$$F_Q = \text{Tr} \rho L^2 = 4t_s^2 (\text{Tr} G^2 - (\text{Tr} G)^2) = 4\gamma^2 t_s^2$$

Here, $G = \gamma(S_z \otimes I + I \otimes S_z)$, which has non-zero eigenvalues on the eigenbasis $|00\rangle, |11\rangle$.

Post-Selection: QFI under Dephasing

Since G only has non-zero eigenvalues in the $|00\rangle, |11\rangle$ subspace, we restrict to this subspace. Consider an entangled state $|\psi\rangle = a|00\rangle + b|11\rangle$. Under the dephasing channel, the off-diagonal elements decay as

$$\eta = e^{-2\tau}, \text{ where } \tau = \left(\frac{t_s}{T_2^*} \right)^p$$

and the phase accumulates as $\phi = \theta + 2\gamma B t_s$. The resulting density matrix becomes

$$\rho = \begin{pmatrix} a^2 & ab\eta e^{i\phi} \\ ab\eta e^{-i\phi} & b^2 \end{pmatrix}$$

In this two-dimensional subspace, we can treat it as a single-qubit system and decompose the density matrix via the Bloch vector

$$\vec{r} = (2ab\eta \cos \phi, 2ab\eta \sin \phi, a^2 - b^2).$$

Post-Selection: QFI under Dephasing

For a single-qubit mixed state, the QFI with respect to B is given by $F_Q^B = |\partial_B \vec{r}|^2 + \frac{(\vec{r} \cdot \partial_B \vec{r})^2}{1 - |\vec{r}|^2}$. Since ∂_B only affects the phase ϕ , the radial component $\vec{r} \cdot \partial_B \vec{r} = 0$, and we obtain

$$F_Q^B = |\partial_B r|^2 = 16\gamma^2 t_s^2 e^{-4\tau} a^2 b^2 \leq 4\gamma^2 t_s^2 e^{-4\tau}$$

Note that in a single-qubit system the maximum QFI is $4\gamma^2 t_s^2 e^{-2\tau}$.

Since quantum Fisher information is additive, a two-qubit separable state is (twice) better than an entangled state.

Post-Selection: Motivation

In this sense, we hypothesize that adding a post-selection procedure might be advantageous. Intuitively, the information on the B -field is encoded in the phase, and through post-selection we may discard unnecessary information to the ancillary qubit (or the other energy level).

Post-Selection: Motivation

In this sense, we hypothesize that adding a post-selection procedure might be advantageous. Intuitively, the information on the B -field is encoded in the phase, and through post-selection we may discard unnecessary information to the ancillary qubit (or the other energy level).

Post-Selection: Separable Filter Ansatz

We prepared the following circuit ansatz.

- State Preparation (Red): To prepare an arbitrary two-qubit state, we employ one CNOT gate.
- Sensing (Orange): The system interacts with the (perturbed) magnetic field δB during sensing time t_s , and the noise channel is embedded.
- Post-Selection and Measurement (Purple): Unitary operations U , V change the post-selection basis and measurement basis.

Post-Selection: Separable Filter Results

Given the learning curve, the optimization appears to have saturated. However, the saturated value was twice the QFI of the post-selected single-qubit system.

Post-Selection: Separable State Verification

The prepared state turns out to be separable, as verified by the negativity of the prepared state.

Post-Selection: Entangled Filter Ansatz

GH suggested enabling an entangled filter and entangled-basis measurement. We modified the circuit ansatz accordingly. A two-qubit gate is added so that post-selection is effectively performed in an entangled basis.

$$K_{\text{eff}} = U(K_1 \otimes K_2)U^\dagger$$

Here, we employ the KAK decomposition (equivalent to using 3 CNOT gates per two-qubit gate) for U , V , in order to explore the full $\text{SU}(4)$ space.

Post-Selection: Entangled Filter Results

Interestingly, the optimized QFI (CFI) exceeds that of the post-selected separable state by a factor of two.

Post-Selection: Entangled Filter Entanglement

The prepared state was separable, but entanglement was recovered after post-selection. Note that the entanglement strength was not maximal; the negativity of a Bell state is 0.5.

Post-Selection: Filter Strength Analysis

Furthermore, the filter strengths become asymmetric, with one post-selection strength γ approaching 1, indicating a strong measurement.

Post-Selection: Fixed Filter Design

I am currently investigating optimal filter design. Under the guidance of Dr. Lee, I fixed the post-selection filter as $K = \sqrt{1 - \gamma}|00\rangle\langle 00| + |11\rangle\langle 11|$. The poor performance is attributed to $[\mathcal{H}_B, K] = 0$.

Post-Selection: X-Basis Filter Design

Next, I tried to maximize the Frobenius norm of the commutator. $[\mathcal{H}_B, K] = 0$,
 $K = \sqrt{1 - \gamma} |++\rangle\langle ++| + |--\rangle\langle --|$.

Post-Selection: Summary and Next Steps

- The message is clear: under dephasing noise, entanglement is vulnerable to noise rather than serving as a resource (the well-known Heisenberg limit). Post-selection onto an entangled basis can recover entanglement and yield a gain in sensing.
- Further analysis is required: prepared state, filter strength, and post-selection/measurement basis.
- I need to write a draft. The target deadline is Apr 09.
- Based on the insights obtained, I am building an analytical framework.

DDrf: Work Flow

1. Comparative Study between CPMG and DDrf (Theory/Numeric)
 - **[Completed] Poster work** ← (Hun)
2. Enhanced (Hybrid) DDrf Gate (Theory/Numeric)
 - [Completed] Tried many ideas... ← (J. J)
 - [Pending?] Alternating Ω_{RF} for odd- and even-numbered pulses ← (J. J)
 - [Ongoing] Draw phase diagram (J. J)
 - **[Ongoing] Analytical Study; Conditional Rotation Angle** ←(Hun)
3. DDrf Spectroscopy (Experiment)
 - [Completed] Numerical Simulation based on Taminiau Paper ← (Hun)
 - [Ongoing] Experiment ← (Dr. Lee)
 - **[Ongoing] Numerical Simulation; side-peak problem** ←(Hun)
4. Multi-qubit Control (Numeric/Experiment)
 - [Pending] ω_0 **control** ←(Hun)

DDrf: QISK Poster

I managed to finish the QISK Poster Presentation
(26/02/25). 🙌

DDrf: Hamiltonian Engineering

DDrf refers to selective, phase-controlled radio-frequency (RF) driving of nuclear spins interleaved with dynamic decoupling (DD) sequences on an electron spin.

I would like to explain DDrf in terms of Hamiltonian engineering.

If we can write the Hamiltonian in a block-diagonal form,

$$H = |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1$$

then the time evolution of the second qubit becomes contingent on the first qubit. For example, if we prepare the first qubit in the $|+\rangle$ state and perform time evolution for t_1 , we have

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow[t_1]{} \frac{1}{\sqrt{2}}(|0\rangle \otimes e^{-iH_0 t_1}|0\rangle + |1\rangle \otimes e^{-iH_1 t_1}|0\rangle)$$

DDrf: Conditional Gate via π -Pulse

If we change the first qubit state using a π -pulse ($|0(1)\rangle \rightarrow |1(0)\rangle$) and apply time evolution for t_2 , then

$$\begin{aligned} |\psi\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) \xrightarrow[t_1]{} \frac{1}{\sqrt{2}}(|0\rangle \otimes e^{-iH_0 t_1}|0\rangle + |1\rangle \otimes e^{-iH_1 t_1}|0\rangle) \\ &\xrightarrow[\pi]{} \frac{1}{\sqrt{2}}(|0\rangle \otimes e^{-iH_1 t_1}|0\rangle + |1\rangle \otimes e^{-iH_0 t_1}|0\rangle) \\ &\xrightarrow[t_2]{} \frac{1}{\sqrt{2}} \left(|0\rangle \otimes \underbrace{e^{-iH_0 t_2} e^{-iH_1 t_1}}_{=U_0} |0\rangle + |1\rangle \otimes \underbrace{e^{-iH_1 t_2} e^{-iH_0 t_1}}_{=U_1} |0\rangle \right) \end{aligned}$$

Generally, $U_0 \neq U_1$, hence it is a conditional gate!

DDrf: Pulse Sequence

DDrf Spectroscopy: Hamiltonian

Given the NV- ^{13}C Hamiltonian, we have

$$\begin{aligned}\mathcal{H} &= \gamma_c B_z I_z^i + A_{||}^i S_z I_z^i + A_{\perp} S_z I_x^i \\ \rightarrow \mathcal{H} &= |0\rangle\langle 0| \otimes \omega_0 I_z + |-1\rangle\langle -1| \otimes (\omega_0 I_z - A_{||} I_z - A_{\perp} I_x)\end{aligned}$$

Then, with additional rf driving, we have

$$\begin{aligned}H &= |0\rangle\langle 0| \otimes H_0 + |1\rangle\langle 1| \otimes H_1 \\ H_0 &= \omega_0 I_z + 2\Omega_{\text{RF}} \cos(\omega_{\text{RF}} t + \phi) I_x \\ H_1 &= \omega_1 \tilde{I}_z + 2\Omega_{\text{RF}} \cos \beta \cos(\omega_{\text{RF}} t + \phi) \tilde{I}_x + 2\Omega_{\text{RF}} \sin \beta \cos(\omega_{\text{RF}} t + \phi) \tilde{I}_z\end{aligned}$$

where

$$\begin{aligned}\omega_0 &= \gamma_c B_z & \cos \beta &= \frac{\omega_0 - A_{\perp}}{\omega_1} & \tilde{I}_z &= \cos \beta I_z + \sin \beta I_x \\ \omega_1 &= \sqrt{(\omega_0 - A_{||})^2 + A_{\perp}^2} & \sin \beta &= \frac{A_{\perp}}{\omega_1} & \tilde{I}_x &= \cos \beta I_x - \sin \beta I_z\end{aligned}$$

DDrf Spectroscopy: Rotating Frame

Two rotating frames are used with respect to the electron spin state:

$$R_0(t) = e^{i\omega_{\text{RF}}tI_z} \quad R_1(t) = e^{i\omega_{\text{RF}}t\tilde{I}_z}$$

In the rotating frame, each Hamiltonian becomes:

$$\begin{aligned} H_0 \rightarrow H'_0 &= R_0(t)(H_0 - \omega_{\text{RF}}I_z)R_0(t)^\dagger \\ &= (\omega_0 - \omega_{\text{RF}})I_z + \Omega_{\text{RF}}(\cos \phi I_x + \sin \phi I_y) \end{aligned}$$

$$\begin{aligned} H_1 \rightarrow H'_1 &= R_1(t)(H_1 - \omega_{\text{RF}}I_z)R_1(t)^\dagger \\ &= (\omega_1 - \omega_{\text{RF}})\tilde{I}_z + \Omega_{\text{RF}} \cos \beta (\cos \phi \tilde{I}_x + \sin \phi \tilde{I}_y) \\ &= (\omega_1 - \omega_{\text{RF}})(\cos \beta I_z + \sin \beta I_x) + \Omega_{\text{RF}} \cos \beta (\cos \beta \cos \phi I_x + \sin \phi I_y - \sin \beta \cos \phi I_z) \end{aligned}$$

In each rotating frame, the time evolution can be readily calculated as $e^{-iH_{0(1)}t}$.

DDrf Spectroscopy: Full Time Evolution

Then, over the full time evolution, the unitary operation can be calculated as:

$$\begin{aligned} U &= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1 \\ U_0 &= R_0(4N\tau)^\dagger e^{-iH'_0\tau} R_0((2N-1)\tau) R_1((2N-1)\tau)^\dagger e^{-iH'_1 2\tau} R_1((2N-3)\tau) R_0((2N-3)\tau)^\dagger e^{-iH'_0\tau} \dots \\ &\quad \dots e^{-iH'_0\tau} R_0(3\tau) R_1(3\tau)^\dagger e^{-iH'_1 2\tau} R_1(\tau) R_0(\tau)^\dagger e^{-iH'_0\tau} R_0(0) \\ U_1 &= R_1(4N\tau)^\dagger e^{-iH'_1\tau} R_1((2N-1)\tau) R_0((2N-1)\tau)^\dagger e^{-iH'_0 2\tau} R_0((2N-3)\tau) R_1((2N-3)\tau)^\dagger e^{-iH'_1\tau} \dots \\ &\quad \dots e^{-iH'_1\tau} R_1(3\tau) R_0(3\tau)^\dagger e^{-iH'_0 2\tau} R_0(\tau) R_1(\tau)^\dagger e^{-iH'_1\tau} R_1(0) \end{aligned}$$

This approach is exact except for the assumption that the MW pulse duration for changing the electron state is negligibly short. It is worth mentioning that

- $\Omega_{\text{RF}} \rightarrow 0, \omega_{\text{RF}} = \omega_1$: CPMG at a certain $\tau \simeq \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$
- $\beta \rightarrow 0, \omega_{\text{RF}} = \omega_1 > \omega_0$: DDrf(2019)
- $\Omega_{\text{RF}} \rightarrow \frac{\Omega_{\text{RF}}}{\cos \beta}$: Jiwon's idea
- $\tau = \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$: Hybrid DDrf

DDrf Spectroscopy: Procedure

The paper [Phys. Rev. X **9**, 031045 (2019)] showed that the DDrf gate offers the additional benefit of detecting spins with small $A_{\perp}$.

Procedure:

1. $\frac{\pi}{2}$ -pulse rotates electron spin to $|+\rangle$.
2. DDrf Gate with fixed N and τ , resulting in $U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1$
3. $\frac{\pi}{2}$ -pulse is applied to electron spin with varying phase ϕ .
 - In our experiment, we measure P_x , the projection onto $|+\rangle$ ($\phi = \frac{\pi}{2}$).

DDrf Spectroscopy: Results

Results:

1. The expectation value becomes $P_x = \frac{1}{2} + \frac{1}{4}\Re(\text{Tr}U_0U_1^\dagger)$
2. Extended to an N -qubit simulation, $P_x = \frac{1}{2} + \frac{1}{2^{N+1}}\Re(\text{Tr}U_0U_1^\dagger)$ where

$$\text{Tr}U_0U_1^\dagger = \prod_{i=1}^N \text{Tr}U_0^iU_1^{i\dagger}.$$

3. In the Taminiau paper, the amplitude is $A = \frac{1}{2^{N+1}}|\text{Tr}U_0U_1^\dagger|$.
4. At the resonant frequency ($\omega_{\text{RF}} = \omega_1$), peaks are observed.
5. Peaks also appear at off-resonant conditions ($\omega_{\text{RF}} = \omega_1 + \frac{2\pi m}{\tau}$, $m \in \mathbb{Z}$) due to the same phase accumulation.

DDrf Spectroscopy: Taminiau Results

DDrf Spectroscopy: Reproduced Results

DDrf Spectroscopy: Limiting Cases

Given the following time evolution

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1$$

$$U_0 = R_0(4N\tau)^\dagger e^{-iH'_0\tau} R_0((2N-1)\tau) R_1((2N-1)\tau)^\dagger e^{-iH'_1 2\tau} R_1((2N-3)\tau) R_0((2N-3)\tau)^\dagger e^{-iH'_0\tau} \dots \\ \dots e^{-iH'_0\tau} R_0(3\tau) R_1(3\tau)^\dagger e^{-iH'_1 2\tau} R_1(\tau) R_0(\tau)^\dagger e^{-iH'_0\tau} R_0(0)$$

$$U_1 = R_1(4N\tau)^\dagger e^{-iH'_1\tau} R_1((2N-1)\tau) R_0((2N-1)\tau)^\dagger e^{-iH'_0 2\tau} R_0((2N-3)\tau) R_1((2N-3)\tau)^\dagger e^{-iH'_1\tau} \dots \\ \dots e^{-iH'_1\tau} R_1(3\tau) R_0(3\tau)^\dagger e^{-iH'_0 2\tau} R_0(\tau) R_1(\tau)^\dagger e^{-iH'_1\tau} R_1(0)$$

- $\Omega_{\text{RF}} \rightarrow 0, \omega_{\text{RF}} = \omega_1$: CPMG at a certain $\tau \simeq \frac{(2k-1)\pi}{2\omega_0 + A_{\parallel}}$
- $\beta \rightarrow 0$, (then $I_z = \tilde{I}_z$), $\omega_{\text{RF}} = \omega_1$: DDrf(2019)

Here, the change of frame is responsible for the CPMG effect at a certain τ , while additional RF driving resides in H_0 and H_1 . However, finding $U_{(0,1)} = e^{-i\theta\hat{\sigma}_{(0,1)}}$ is not straightforward...

DDrf: Per-Cell Propagator

Note that the DDrf sequence is built up recursively from MW and RF pulses. Each building block (train) $V^{(k)}$ constitutes a unitary operation. In short, $V^{(k)} = \mathcal{T} \left(\exp -i \int \mathcal{H} dt \right)$.

$$V^{(k)} = |0\rangle\langle 0| \otimes V_0^{(k)} + |1\rangle\langle 1| \otimes V_1^{(k)}$$

DDrf: Total Unitary Decomposition

Then, the total unitary operation becomes:

$$U = |0\rangle\langle 0| \otimes \prod_{k=1}^{N/2} V_0^{(k)} + |1\rangle\langle 1| \otimes \prod_{k=1}^{N/2} V_1^{(k)}.$$

This equation is equivalent to the previous one, but implies the following strategy. The idea is that any unitary operation can be written as e^{-iA} for an arbitrary Hermitian operator A . At the same time, it can be decomposed into $e^{-iA_1}e^{-iA_2}$, but this decomposition is non-trivial and not unique. This means we can choose a favorable form to evaluate the equation above without fully computing the previous expression.

DDrf: Taminiau Limit ($\beta \rightarrow 0$)

For example, in the $\beta \rightarrow 0$, $\omega_{\text{RF}} = \omega_1$ case (Taminiau, 2019),

$$\begin{aligned} V_0^k &= e^{-iH_0\tau} e^{-iH_1 2\tau} e^{-iH_0\tau} \\ &= e^{-i\delta_0\tau I_z} e^{-i2\Omega\tau \hat{\phi}_{2k}} e^{-i\delta_0\tau I_z} \\ &= \left[e^{-i\delta_0\tau I_z} e^{-i\delta_0\tau I_z} \right] \left[e^{+i\delta_0\tau I_z} e^{-i2\Omega\tau \hat{\phi}_{2k}} e^{-i\delta_0\tau I_z} \right] \\ &= e^{-i\delta_0 2\tau I_z} e^{-2\Omega\tau \hat{\phi}'_{2k}} \end{aligned}$$

where $\hat{\phi} = \cos \phi I_x + \sin \phi I_y$ and $\hat{\phi}'_{2k} = \cos(\phi - \delta\tau) I_x + \sin(\phi - \delta\tau) I_y$.

DDrf: Successive Products

In successive products, we can continue applying this trick.

$$\begin{aligned} V_0^k V_0^{(k-1)} &= e^{-i\delta_0 2\tau I_z} e^{-2\Omega\tau \hat{\phi}'_{2k}} e^{-i\delta_0 2\tau I_z} e^{-2\Omega\tau \hat{\phi}'_{2k-2}} \\ &= \left[e^{-i\delta_0 2\tau I_z} e^{-i\delta_0 2\tau I_z} \right] \left[e^{+i\delta_0 2\tau I_z} e^{-2\Omega\tau \hat{\phi}'_{2k}} e^{-i\delta_0 2\tau I_z} \right] e^{-2\Omega\tau \hat{\phi}'_{2k-2}} \\ &= e^{-i\delta_0 4\tau I_z} e^{-2\Omega\tau \hat{\phi}''_{2k}} e^{-2\Omega\tau \hat{\phi}'_{2k-2}} \end{aligned}$$

By setting $\phi_{2k} = (2k - 1)\delta\tau + \phi_0$, we align all ϕ_{2k} to the same axis, enabling successive conditional rotation buildup.

DDrf: Taylor Expansion in β

Now, in the exact time-evolution equation, we can define V_0 and V_1 for each cell.

$$\begin{aligned} V_0^{(k)} &= e^{-iH_0\tau} R_0((4k-3)\tau) R_1^\dagger((4k-3)\tau) e^{-iH_1 2\tau} R_1((4k-1)\tau) R_0^\dagger e^{-iH_0\tau} \\ V_1^{(k)} &= e^{-iH_1\tau} R_1((4k-3)\tau) R_0^\dagger((4k-3)\tau) e^{-iH_0 2\tau} R_0((4k-1)\tau) R_1^\dagger e^{-iH_1\tau} \end{aligned}$$

To identify the CPMG effect by choosing a proper τ , I employ a Taylor-expansion-like approach. That is,

$$\begin{aligned} V_0^{(k)} &\simeq V_0^{(k)} \Big|_{\beta=0} + \beta \frac{\partial}{\partial \beta} V_0^{(k)} \Big|_{\beta=0} + \mathcal{O}(\beta^2) \\ V_1^{(k)} &\simeq V_1^{(k)} \Big|_{\beta=0} + \beta \frac{\partial}{\partial \beta} V_1^{(k)} \Big|_{\beta=0} + \mathcal{O}(\beta^2) \end{aligned}$$

Under this approach, we expect to find a form $e^{-i\alpha I_z} e^{-i\theta_\beta \hat{\sigma}_\beta}$ where $\alpha, \theta_\beta, \hat{\sigma}_\beta$ are arbitrary parameters and axes, and β denotes β -dependence. Furthermore, we can **verify this approximation by testing its limits at $\beta \rightarrow 0$ and $\Omega \rightarrow 0$** , which correspond to Taminiau DDrf and CPMG, respectively.

DDrf: Perturbative Calculation

Without showing the full derivation (assuming $\delta_1 \ll \Omega \ll \delta_0$), I obtained

$$\begin{aligned}
 V_0^{(k)} = & e^{-i2\delta_0\tau I_z} \left[\underbrace{\left(\cos \Omega\tau + \frac{\beta}{2} \sin \Omega\tau \cos(\phi_{2k} + (4k-2)\omega_{\text{RF}}\tau) \right)}_{=\cos \theta} I \right. \\
 & + -i \left(\sin \Omega\tau \hat{\phi}' + \beta \left(\cos \omega_{\text{RF}}\tau \sin \Omega\tau \cos(\phi_{2k} + (4k-2)\omega_{\text{RF}}\tau) I_z \right. \right. \\
 & \left. \left. - \sin \omega_{\text{RF}}\tau \cos \Omega\tau (\cos((4k-2)\omega_{\text{RF}}\tau) I_x - \sin((4k-2)\omega_{\text{RF}}\tau) I_y) \right) \right] \\
 V_1^{(k)} = & e^{-i2\delta_0\tau I_z} \left[e^{-iH_1(\phi'_{2k+1})\tau} e^{-iH_1(\phi_{2k-1})\tau} + i\beta \sin \delta_0\tau e^{-iH_1(\phi'_{2k+1})\tau} \hat{\chi} e^{-iH_1(\phi_{2k-1})\tau} \right]
 \end{aligned}$$

where $\hat{\chi} = \cos((4k-2)\omega\tau + \delta_0\tau)I_x - \sin((4k-2)\omega\tau + \delta_0\tau)I_y$. Since the $V_1^{(k)}$ term involves two phases, I did not proceed to compute the full expression. Optimistically, one might choose appropriate τ and ϕ_k such that $\theta > \Omega\tau$ with antiparallel rotation axes, enabling successive conditional rotation buildup.

DDrf Spectroscopy: Side-Peak Problem

In the Taminiau paper, given τ , Ω_{RF} , N , and $\omega_{\text{RF}} = \omega_1$, the DDrf gate operation is

$$U = |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1$$

where:

$$\begin{aligned} U_0 &= R_z(N(\omega_L - \omega_1)\tau) \cdot R_\phi(N\Omega_{\text{RF}}\tau) \\ U_1 &= R_z(N(\omega_L - \omega_1)\tau) \cdot R_\phi(-N\Omega_{\text{RF}}\tau) \end{aligned}$$

where $R_z(\theta) = e^{-i\theta I_z}$ and $R_\phi(\theta) = e^{-i\theta(\cos \phi I_x + \sin \phi I_y)}$.

[Observation] The operation becomes unconditional when $N\Omega_{\text{RF}}\tau = 2\pi$; a flat spectroscopy signal was expected.

DDrf Spectroscopy: Side-Peak Problem



DDrf Spectroscopy: Side-Peak Problem



DDrf Spectroscopy: Side-Peak Problem



DDrf Spectroscopy: Side-Peak Problem

[**Analysis**] DDrf spectroscopy can be understood from the perspective of Rabi oscillations. Although an analytical form of Ω_{eff} is difficult to derive (though $\Omega_{\text{eff}} \propto \Omega_{\text{RF}}$), when the RF frequency is detuned from resonance by $\delta = \omega_1 - \omega_{\text{RF}}$, the generalized Rabi frequency is

$$\Omega_{\text{gen}} = \sqrt{\delta^2 + \Omega_{\text{eff}}^2}$$

The signal may take the form:

$$P_x \simeq 1 - \underbrace{\frac{\Omega_{\text{eff}}}{\Omega_{\text{gen}}}}_{\text{Lorentzian envelope}} \underbrace{\sin^2 \frac{\Omega_{\text{gen}} 2N\tau}{2}}_{\text{oscillation}}$$

DDrf: Detuned Rotating Frame

We have observed that the side-peak problem arises near the resonant peak $\omega_{\text{RF}} \simeq \omega_1$. Let us return to the rotating-frame Hamiltonian.

$$\begin{aligned} H_0 &\rightarrow H'_0 = R_0(t)(H_0 - \omega_{\text{RF}}I_z)R_0(t)^\dagger \\ &= \delta_0 I_z + \Omega_{\text{RF}}(\cos \phi I_x + \sin \phi I_y) \\ H_1 &\rightarrow H'_1 = R_1(t)(H_1 - \omega_{\text{RF}}I_z)R_1(t)^\dagger \\ &= \delta_1 \tilde{I}_z + \Omega_{\text{RF}} \cos \beta (\cos \phi \tilde{I}_x + \sin \phi \tilde{I}_y) \end{aligned}$$

where $\delta_{(0,1)} = \omega_{0(1)} - \omega_{\text{RF}}$. Here, let us assume $\hat{\tilde{I}} = \hat{I}$ for simplicity, i.e., $A_\perp \simeq 0$ and $\beta \rightarrow 0$.

Previously, we assumed $\delta_1 \ll \Omega \ll \delta_0$. To see the detuned effect, let us keep δ_1 small but not negligible.

DDrf: Tilted-Axis Propagator

With $H_1(\phi) = \delta_1 I_z + \Omega \hat{\phi} \cdot \vec{I}$, the propagator $e^{-iH_1 t}$ is now a rotation about a **tilted** axis. Define

$$\Omega_{\text{eff}} = \sqrt{\Omega^2 + \delta_1^2}, \quad \sin \gamma = \frac{\delta_1}{\Omega_{\text{eff}}}, \quad \cos \gamma = \frac{\Omega}{\Omega_{\text{eff}}}$$

so that

$$H_1(\phi) = \Omega_{\text{eff}} \hat{n}(\phi) \cdot \vec{I}, \quad \hat{n}(\phi) = (\cos \gamma \cos \phi, \cos \gamma \sin \phi, \sin \gamma).$$

The axis $\hat{n}(\phi)$ is tilted out of the xy -plane by angle γ .

DDrf: Phase Telescoping with Detuning

The conjugation trick still works, since $e^{i\alpha I_z}(\hat{n} \cdot \vec{I})e^{-i\alpha I_z}$ only rotates the azimuthal angle by $-\alpha$ and leaves the z -component ($\sin \gamma$) invariant. So exactly the same manipulations you wrote give:

$$V_0^{(k)} = e^{-i2\delta_0\tau I_z} e^{-i\Omega_{\text{eff}} 2\tau} \hat{n}(\phi'_{2k}) \cdot \vec{I}$$

with $\phi'_{2k} = \phi_{2k} - \delta_0\tau = (2k - 2)\delta_0\tau$, and

$$V_1^{(k)} = e^{-i2\delta_0\tau I_z} e^{-i\Omega_{\text{eff}} 2\tau} \hat{n}(\phi_{2k-1}) \cdot \vec{I}$$

with $\phi_{2k-1} = (2k - 2)\delta_0\tau + \pi$. The phase protocol ensures the two RF pulses within each V_1 cell collapse onto the same tilted axis, just as before.

DDrf: Total Propagator with Detuning

Composing over $N/2$ cells, each factor of $e^{-i2\delta_0\tau I_z}$ shifts subsequent azimuthal angles by $2\delta_0\tau$, which exactly compensates the k -dependent phase stepping. The tilt angle γ is k -independent, so it passes through the telescoping unchanged:

$$V_0^{\text{tot}} = e^{-iN\delta_0\tau I_z} e^{-i\Omega_{\text{eff}} N\tau \hat{n}_0 \cdot \vec{I}}, \quad \hat{n}_0 = (\cos \gamma, 0, \sin \gamma)$$

$$V_1^{\text{tot}} = e^{-iN\delta_0\tau I_z} e^{-i\Omega_{\text{eff}} N\tau \hat{m}_0 \cdot \vec{I}}, \quad \hat{m}_0 = (-\cos \gamma, 0, \sin \gamma)$$

The RZ factor $e^{-iN\delta_0\tau I_z}$ is the same as the $\delta_1 = 0$ case, so the interesting physics is in the rotations. Comparing the two axes:

$$\hat{n}_0 = \cos \gamma \hat{x} + \sin \gamma \hat{z}, \quad \hat{m}_0 = -\cos \gamma \hat{x} + \sin \gamma \hat{z}.$$

DDrf: Full Conditional Unitary

Full unitary

Defining $\alpha \equiv N\delta_0\tau$ and $\theta \equiv \Omega_{\text{eff}} N\tau$:

$$U = (\mathbb{1}_e \otimes e^{-i\alpha I_z}) \cdot \left[|0\rangle\langle 0| \otimes e^{-i\theta \hat{n}(0) \cdot \vec{I}} + |1\rangle\langle 1| \otimes e^{-i\theta \hat{n}(\pi) \cdot \vec{I}} \right]$$

The first factor is an **unconditional** nuclear z -rotation (identical for both electron states). The second factor is the **conditional gate**.

DDrf: Detuned Rabi Formula

Substituting $\Omega_{\text{eff}}^2 = \Omega^2 + \delta_1^2$ and $\sin^2 \gamma = \delta_1^2 / (\Omega^2 + \delta_1^2)$:

$$\text{Tr}(U_0 U_1^\dagger) = 2 \cos\left(\sqrt{\Omega^2 + \delta_1^2} N\tau\right) + \frac{4\delta_1^2}{\Omega^2 + \delta_1^2} \sin^2\left(\frac{\sqrt{\Omega^2 + \delta_1^2} N\tau}{2}\right)$$

Using $2 \cos \theta = 2 - 4 \sin^2(\theta/2)$:

$$\frac{1}{2} \text{Tr}(U_0 U_1^\dagger) = 1 - \frac{2\Omega^2}{\Omega^2 + \delta_1^2} \sin^2\left(\frac{\sqrt{\Omega^2 + \delta_1^2} N\tau}{2}\right)$$

This is exactly the **detuned Rabi formula**. Yay!

DDrf: Taylor Expansion of Trace

Given $\frac{\delta_0}{\Omega} \ll 1$, applying a Taylor expansion up to first order, we have

$$\frac{1}{2} \text{Tr}(U_0 U_1^\dagger) = 1 - \underbrace{2\Omega^2 \left(1 - \frac{\delta_0^2}{\Omega^2}\right)}_{\text{Lorentzian envelope}} \underbrace{\sin^2 \left(N\tau\delta_0^2 + \frac{N\Omega\tau}{2}\right)}_{\text{sidelobe}}$$

We can see:

- The Lorentzian envelope takes a quadratic form.
- The oscillation produces sidelobes with a period of $\frac{2\pi}{N\tau}$.

DDrf: Comparison with Numerical Result

It seems that this equation explains the curve very well!

$$\frac{1}{2}\text{Tr}(U_0 U_1^\dagger) = 1 - 2\Omega^2 \left(1 - \frac{\delta_0^2}{\Omega^2}\right) \sin^2 \left(N\Omega\delta_0^2 + \frac{N\Omega\tau}{2}\right)$$

DDrf: Apodized Pulse Shaping

Per-cell propagators with $\Omega_k = \Omega f(k)$

Although it may be unrealistic, the idea is to use apodized pulse RF driving like this...

DDrf: Apodized Pulse Shaping

And it works(?).

DDrf: Apodized Pulse Shaping

DDrf: Per-Cell Propagators with Window

The per-cell propagators are:

$$V_0^{(k)} = e^{-i2\delta_0\tau I_z} e^{-i\theta_k \hat{n}_k(0) \cdot \vec{I}}, \quad V_1^{(k)} = e^{-i2\delta_0\tau I_z} e^{-i\theta_k \hat{n}_k(\pi) \cdot \vec{I}}$$

with k -dependent quantities:

$$\theta_k = 2\Omega_{\text{eff},k} \tau, \quad \Omega_{\text{eff},k} = \sqrt{\Omega_k^2 + \delta_1^2}, \quad \sin \gamma_k = \frac{\delta_1}{\Omega_{\text{eff},k}}, \quad \cos \gamma_k = \frac{\Omega_k}{\Omega_{\text{eff},k}}$$

$$\hat{n}_k(0) = (\cos \gamma_k, 0, \sin \gamma_k), \quad \hat{n}_k(\pi) = (-\cos \gamma_k, 0, \sin \gamma_k)$$

Note: the tilt angle γ_k now **varies with k** because Ω_k does.

DDrf: Telescoping with Window

Commuting $e^{-i2\delta_0\tau I_z}$ through R_k shifts the azimuthal angle by $-2\delta_0\tau$ while preserving γ_k and θ_k . By induction (exactly as before):

$$\prod_{k=1}^{N/2} V_0^{(k)} = e^{-iN\delta_0\tau I_z} \prod_{k=N/2}^1 e^{-i\theta_k \hat{n}_k(0) \cdot \vec{I}},$$

$$\prod_{k=1}^{N/2} V_1^{(k)} = e^{-iN\delta_0\tau I_z} \prod_{k=N/2}^1 e^{-i\theta_k \hat{n}_k(\pi) \cdot \vec{I}}$$

Crucial difference from the constant- Ω case: the ordered product does **not** collapse into a single rotation, because $\hat{n}_k(0)$ has k -dependent tilt γ_k , and rotations about different axes in the xz -plane do not commute and are not aligned anymore.

Exception — $\delta_1 = 0$: all $\gamma_k = 0$, all axes align ($\hat{x}$ or $-\hat{x}$), and the products do collapse:

$$\prod_{k=N/2}^1 e^{-i \cdot 2\Omega_k \tau I_x} = e^{-i\Theta I_x}, \quad \Theta \equiv 2\Omega\tau \sum_{k=1}^{N/2} f(k)$$

DDrf: Clean Windowed Result

Case $\delta_1 = 0$: clean windowed result

All V_0 rotations are about $+\hat{x}$, all V_1 rotations about $-\hat{x}$:

$$\text{tr}(U_0 U_1^\dagger)|_{\delta_1=0} = \text{tr}(e^{-i\Theta I_x} \cdot e^{-i\Theta I_x}) = \text{tr}(e^{-i2\Theta I_x}) = 2 \cos \Theta$$

$$\boxed{\text{tr}(U_0 U_1^\dagger)|_{\delta_1=0} = 2 \cos \left(2\Omega\tau \sum_{k=1}^{N/2} f(k) \right)}$$

DDrf: Perturbative Result

Case $\delta_1 \neq 0$: perturbative result

For small δ_1 , define the zeroth-order (unperturbed, $\delta_1 = 0$) rotation and the perturbation:

$$e^{-i\theta_k \hat{n}_k(0) \cdot \vec{I}} = e^{-i(2\Omega_k \tau I_x + 2\delta_1 \tau I_z)} + \mathcal{O}(\delta_1^2)$$

where I used $\theta_k \cos \gamma_k = 2\Omega_k \tau + \mathcal{O}(\delta_1^2)$ and $\theta_k \sin \gamma_k = 2\delta_1 \tau$ (exactly, independent of k).

Going to the interaction picture with respect to the I_x rotations, define the cumulative angle after cell j :

$$\Phi_j \equiv 2\Omega \tau \sum_{m=1}^j f(m)$$

DDrf: Interaction Picture

The I_z perturbation in cell k is rotated into the frame where cells $1, \dots, k-1$ have already been applied:

$$I_z \longrightarrow \cos \Phi_{k-1} I_z - \sin \Phi_{k-1} I_y$$

Summing the first-order contribution over all cells:

$$\prod_{k=N/2}^1 e^{-i\theta_k \hat{n}_k(0) \cdot \vec{I}} \approx e^{-i\Theta I_x} \exp \left(-i2\delta_1 \tau \sum_{k=1}^{N/2} [\cos \Phi_{k-1} I_z - \sin \Phi_{k-1} I_y] \right)$$

Similarly for the V_1 branch (with $I_x \rightarrow -I_x$). When we form $U_0 U_1^\dagger$, the leading $e^{-i\Theta I_x}$ terms combine into $e^{-i2\Theta I_x}$, and the first-order corrections produce:

$$\boxed{\frac{1}{2} \text{tr}(U_0 U_1^\dagger) \approx \cos \Theta - \frac{2\delta_1^2 \tau^2}{\sin^2(\Theta/2)} \sin^2(\Theta/2) \left| \sum_{k=1}^{N/2} f_k e^{i\Phi_{k-1}} \right|^2 \cdot (\dots)}$$

DDrf: Spectral Response

Defining the normalized window:

$$F(\delta_1) \equiv \sum_{k=1}^{N/2} f(k) e^{i\Phi_{k-1}(\delta_1)}$$

the spectral response near resonance takes the form of a (windowed) discrete Fourier transform:

$$\frac{1}{2} \text{tr}(U_0 U_1^\dagger) \approx -1 + \text{const} \times \left| \frac{F(\delta_1)}{F(0)} \right|^2 \cdot \delta_1^2$$

where $F(0) = \sum_k f(k)$ is just the normalization.

DDrf: Window Functions

Each window $f(k)$ has a continuous counterpart $w(t)$, and the spectral response is the normalized Fourier transform squared. All windows in the cosine family can be written as $w(t) = a_0 - a_1 \cos(2\pi t/T) + a_2 \cos(4\pi t/T)$. The normalized spectral response has the universal form:

$$\left| \frac{F(\delta_1)}{F(0)} \right|^2 = \text{sinc}^2(u) \cdot |G(u)|^2$$

where $\text{sinc}(u) = \sin(\pi u)/(\pi u)$ and the **window kernel** $G(u)$ depends on the coefficients.

Rectangular ($a_0 = 1, a_1 = a_2 = 0$): $\left| \frac{F}{F(0)} \right|_{\text{rect}}^2 = \text{sinc}^2(u)$

DDrf: Window Functions

- **Hanning** ($a_0 = a_1 = 1/2, a_2 = 0$): $\left| \frac{F}{F(0)} \right|_{\text{Hann}}^2 = \frac{\text{sinc}^2(u)}{(1-u^2)^2}$

The factor $(1 - u^2)^{-2}$ kills the sinc sidelobes: each sinc zero at integer u is now cancelled by the denominator zero at $u = \pm 1$, leaving only the zero structure at $u = \pm 2, \pm 3, \dots$

- **Hamming** ($a_0 = 0.54, a_1 = 0.46, a_2 = 0$): $\left| \frac{F}{F(0)} \right|_{\text{Hammm}}^2 = \text{sinc}^2(u) \cdot \left(\frac{50u^2 - 27}{27(u^2 - 1)} \right)^2$

The numerator $50u^2 - 27$ has a zero at $u = \sqrt{27/50} \approx 0.735$, which partially cancels the first sinc sidelobe. The imperfect cancellation at $u = 1$ (unlike Hann) gives a small residual, but the first sidelobe is pushed down to -42.7 dB.

- **Blackman** ($a_0 = 0.42, a_1 = 0.5, a_2 = 0.08$): $\left| \frac{F}{F(0)} \right|_{\text{Black}}^2 = \text{sinc}^2(u) \cdot \left(\frac{50u^4 - 209u^2 + 84}{21(u^2 - 1)(u^2 - 4)} \right)^2$

The extra $\cos(4\pi t/T)$ harmonic introduces the $(u^2 - 4)$ denominator, cancelling the sinc zeros at both $u = \pm 1$ and $u = \pm 2$, pushing the first nonzero sidelobe out to $u \approx 3$.

DDrf: Resolution Comparison

Using the π -gate condition $2\Omega\tau \sum_k f(k) = \pi$ with $\sum_k f(k) = M\bar{f}$:

$$u = \frac{\delta_1}{2\Omega\bar{f}}$$

where $\bar{f}$ is the mean of the window. The FWHM in physical units:

Window	$\bar{f}$	FWHM (u)	FWHM (δ_1)
Rectangular	1.00	0.89	1.77 Ω
Hanning	0.50	1.44	1.44 Ω
Hamming	0.54	1.30	1.41 Ω
Blackman	0.42	1.68	1.41 Ω

DDrf: Non-Commutativity Estimate

How badly is it broken?

The mismatch between consecutive axes is

$$\Delta\gamma_k \equiv \gamma_{k+1} - \gamma_k \approx -\frac{\delta_1}{\Omega^2} \Omega \Delta f_k$$

where $\Delta f_k = f(k+1) - f(k)$. So the non-commutativity enters at order $\delta_1 \cdot \Delta f_k$, which is small when either δ_1 is small or the window varies slowly.

DDrf Spectroscopy: Side-Peak Problem

[**Suggestion**] Apply pulse shaping (e.g., a Gaussian envelope).

→ Ω_{RF} becomes time-dependent.

The time-evolution method used here *assumes* Ω_{RF} is time-independent, as required by the RWA.

$$\begin{aligned} U &= |0\rangle\langle 0| \otimes U_0 + |1\rangle\langle 1| \otimes U_1 \\ U_0 &= R_0(4N\tau)^\dagger e^{-iH'_0\tau} R_0((2N-1)\tau) R_1((2N-1)\tau)^\dagger e^{-iH'_1 2\tau} R_1((2N-3)\tau) R_0((2N-3)\tau)^\dagger e^{-iH'_0\tau} \dots \\ &\quad \dots e^{-iH'_0\tau} R_0(3\tau) R_1(3\tau)^\dagger e^{-iH'_1 2\tau} R_1(\tau) R_0(\tau)^\dagger e^{-iH'_0\tau} R_0(0) \\ U_1 &= R_1(4N\tau)^\dagger e^{-iH'_1\tau} R_1((2N-1)\tau) R_0((2N-1)\tau)^\dagger e^{-iH'_0 2\tau} R_0((2N-3)\tau) R_1((2N-3)\tau)^\dagger e^{-iH'_1\tau} \dots \\ &\quad \dots e^{-iH'_1\tau} R_1(3\tau) R_0(3\tau)^\dagger e^{-iH'_0 2\tau} R_0(\tau) R_1(\tau)^\dagger e^{-iH'_1\tau} R_1(0) \end{aligned}$$

Without this scheme, one would need to solve the Schrödinger equation directly, which takes tens of minutes per frequency point. For spectroscopy, I typically sweep hundreds to thousands of ω_{RF} values.

DDrf Spectroscopy: Gaussian Pulse Approach

Nevertheless, if we assume Ω_{RF} changes slowly enough compared to ω_{rf} , the error comes from the RWA. The next problem is to evaluate $e^{-iH'_{0,1}\tau}$ where

$$\Omega_{\text{RF}} = \Omega_0 e^{-\frac{(t-t_k)^2}{2\sigma^2}}.$$

To avoid solving the Schrödinger equation each time, I employ the Magnus expansion.

DDrf Spectroscopy: Magnus Expansion

Magnus expansion

The solution of the differential equation $Y' = A(t)Y$ with initial condition $Y(0) = Y_0$ can be written as $Y(t) = \exp(\Omega(t))$ with $\Omega(t)$ defined by

$$\Omega' = d \exp_{\Omega}^{-1} A(t), \Omega(0) = 0$$

where

$$d \exp_{\Omega}^{-1}(A) = \sum_{k=0}^{\infty} \frac{B_k}{k!} \text{ad}_{\Omega}^k A.$$

In our case, we have the differential equation

$$\frac{\partial}{\partial t} U(t) = -iH(t)U(t).$$

DDrf Spectroscopy: Magnus Expansion

Generally, we try to find the solution in the form of a series

$$\Omega(t) = \sum_{n=1}^{\infty} \Omega_n(t)$$

In this form, there is a well-known solution.

$$\begin{aligned}\Omega_1(t) &= \int_0^t dt_1 A_1 \\ \Omega_2(t) &= \frac{1}{2} \int_0^t dt_1 \int_0^{t_1} dt_2 [A_1, A_2] \\ \Omega_3(t) &= \frac{1}{6} \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 [[A_1, [A_2, A_3]] + [[A_1, A_2], A_3]] \\ \Omega_4(t) &= \frac{1}{12} \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \int_0^{t_3} dt_4 [[[[A_1, A_2], A_3], A_4] + [A_1, [[A_2, A_3], A_4]] + [A_1, [A_2, [A_3, A_4]]] + [A_2, [A_3, [A_4, A_1]]]]\end{aligned}$$

DDrf Spectroscopy: Magnus Expansion

At this point, an important question arises: the solution $e^{\Omega(t)}$ derived from the series $\Omega(t) = \sum_{n=1}^{\infty} \Omega_n(t)$ can be an exact or approximate solution. It is challenging to verify, but at least we can and should verify whether

- $e^{\Omega(t)}$ is in the Lie algebra $\mathfrak{g}$.
- The series converges.

DDrf Spectroscopy: Magnus Expansion Results

Here are the derived solutions:

$$\Omega_1(T) = -i(\delta_{(0,1)} T I_z^{(0,1)} + c_1(\cos \phi I_x^{(0,1)} + \sin \phi I_y^{(0,1)}))$$

$$\Omega_2(T) = 0$$

$$\Omega_3(T) = -i \frac{i\delta_{(0,1)}^2}{24} K_1(\cos \phi I_x^{(0,1)} + \sin \phi I_y^{(0,1)}) - i \frac{i\delta_{0,1}}{24} K_2 I_z^{(0,1)}$$

$$\Omega_4(T) = 0$$

where

$$c_1 = \int_0^T dt f(t)$$

$$K_1 = \int_0^T dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 (2f(t_2) - f(t_1) - f(t_3))$$

$$K_2 = \int_0^T dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 (2f(t_1)f(t_3) - f(t_1)f(t_2) - f(t_2)f(t_3))$$

DDrf: Convergence

Trivially, $e^{\Omega(T)}$ is in the Lie algebra $\mathfrak{g}$. It is known that convergence holds if

$$\int_0^T \|A(s)\|_2 ds < \pi.$$

I believe this may hold, but I have not yet run the simulation to verify.

Thank You!

Questions & Discussion